

TOI-2046b

R-2300

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_250808 · created 2026-08-17T22:57:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460895.934 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.013
Transit depth (Rp/Rs)^2	1.48 +/- 0.32 [%]
Orbital Inclination (inc)	78.62 +/- 3.1
Airmass coefficient 1 (a1)	1.0614 +/- 0.0091
Airmass coefficient 2 (a2)	-0.04 +/- 0.12
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	None
Optimal Aperture	3.36
Optimal Annulus	10.4
Transit Duration (day)	0.05 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.013 vs 0.1213 (deviation 0.03σ)
Duration fit vs published	0.05 d vs 0.1033 d (deviation 0.73σ)
Mid-time O–C	-13.6 min
Depth SNR	4.4
Residual scatter	0.85 %

Light curve

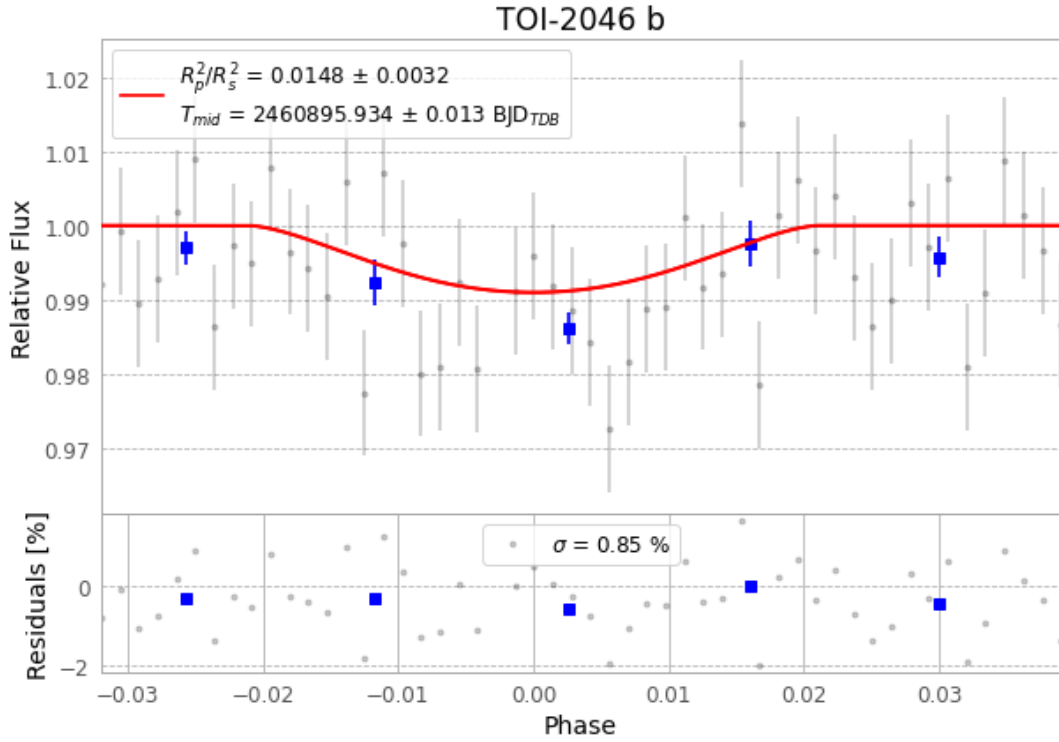


Figure 1 — FinalLightCurve_TOI-2046 b_2025-08-08.png (SHA-256 116100849d4ede65...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 233], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a45ce4c100fb465f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

52 FITS frames (34 MB), TOI2046250808091515.FITS → TOI2046250808114815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-250808.log (028aa38ee709...), FinalLightCurve_TOI-2046 b_2025-08-08.png (116100849d4e...), FinalLightCurve_TOI-2046 b_2025-08-08.csv (1d9f39f7c6b1...)

Compute resources

Wall time 82.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 22:57Z.